

# Near-Eye Purkinje Image Simulation and Capture for Accommodation-Aware Eye Tracking

## Abstract

## CCS Concepts

- Human-centered computing → Mixed / augmented reality;
- Computing methodologies → Rendering; Computer vision.

## Keywords

eye tracking, deep learning, computer graphics

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## 1 Introduction

## 2 Related works

### 2.1 Eye and Gaze Tracking

In stereoscopic and near-eye displays, the **Vergence-Accommodation Conflict (VAC)** arises when the eyes converge to a virtual 3D depth while the crystalline lens remains focused on the physical display plane. This mismatch can reduce fusion performance and cause visual discomfort [Hoffman et al. 2008]. Recent VAC-related depth-perception studies therefore use eye-tracking and controlled optical setups to measure vergence responses under different perceived-depth conditions, highlighting the need for objective and reproducible eye-based measurements in VAC-related experiments [Bontha and Arefin 2024; Sturgeon et al. 2025]. However, vergence alone does not fully describe the accommodative state of the eye. Accommodation is harder to measure directly, especially in compact near-eye systems. Higher-order Purkinje images provide a promising optical cue because they arise from reflections at ocular interfaces, including the crystalline lens surfaces, and vary with the lens geometry during accommodation. Lu et al. [2020] use a multi-camera, multi-LED AR-glasses prototype to track higher-order Purkinje images, extending conventional pupil–corneal-reflection tracking toward accommodation-aware sensing in a near-eye form factor. Ozhan et al. [2023] extend this direction by collecting multi-subject Purkinje measurements and combining a Purkinje imaging setup with a MLP-based regression model to estimate dynamic vergence and accommodation across users.

These works suggest that higher-order Purkinje reflections can serve as physically meaningful cues for accommodation-aware eye tracking, but they also reveal a practical challenge: P3 and P4 are faint, easily defocused, and highly sensitive to LED–camera–eye

**Table 1: Comparison of representative eye-tracking, Purkinje-based accommodation measurement, and near-eye modeling systems.**

| Work                       | Camera / illumination                                            | Mode                                                        | Reported accuracy / result                           |
|----------------------------|------------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------|
| Cornsweet and Crane [1973] | Dual-Purkinje tracker; infrared illumination                     | optical First/fourth Purkinje position tracking             | $\sim 1$ arcmin eye-position accuracy                |
| Morimoto et al. [2002]     | Single camera; at least two NIR light sources                    | Remote gaze tracking with Gullstrand eye model              | $\sim 5^\circ$ gaze error at 500–600 mm              |
| Sturgeon et al. [2025]     | Integrated eye tracker; OST AR display                           | Vergence response measurement under perceived depth changes | EVA response analysis                                |
| Bontha and Arefin [2024]   | Controlled AR setup                                              | Replicated AR haploscope for depth presentation             | Reproducible optical setup                           |
| Ozhan et al. [2023]        | Single NIR camera; one 940 nm NIR LED                            | Purkinje-based vergence and accommodation measurement       | 0.18° vergence / 0.17 D accommodation                |
| Lu et al. [2020]           | Multiple cameras; multiple IR LEDs                               | Near-eye AR Purkinje tracking                               | 0.3782° vergence; 1.108 cm accommodation-depth error |
| RIT-Eyes [2020]            | Synthetic camera; Blender/Cycles illumination                    | Synthetic near-eye image generation                         | 86.06 global mIoU for eye-region segmentation        |
| EyeNeRF [2022]             | Sparse cameras and lights; handheld camera with co-located light | Hybrid mesh-NeRF eye reconstruction and relighting          | Photorealistic eye synthesis / relighting            |
| Khademi et al. [2026]      | Sparse multi-view cameras; point lights/fringe projector         | Relightable mesh-NeRF eye reconstruction                    | Realistic relightable eye reconstruction             |

geometry. Motivated by this limitation, we first build a controllable Blender/LuxCore optical eye model to analyze the formation, visibility, and stability of P1–P4 under near-eye infrared imaging conditions.

### 2.2 Reconstructive surgery of the eyeball with adjustable lighting and end-to-end physical simulation

Excerpt for physical capture, for the optical eyes modeling, the main problem including the complicated geometry of outsurface of cornea. For example, eyenerf propose a hybrid representation, combine explicit eyeball model to handle high-frequency high-gloss, in the mean time, Training the Nerf model for implicit representation enables the modeling of the surrounding eye area and the complex appearance within the eye [Li et al. 2022]. In addition,

*Reconstructing Realistic and Relightable Eyes* highlight the importance of near-field illuminationb to eyes modelling this use hybrid mesh + NeRF Rebuild the ocular system with reflective light capabilities and support the common near-field point lights and structured/fringe projectors [Khademi et al. 2026].

Differentiable optical design provides a useful reference for future optimization of the LED–camera layout in our system. *End-to-end Optimization of Fluidic Lenses* integrates fluidic lens geometry, differentiable ray tracing, and neural reconstruction into a single end-to-end pipeline, allowing the loss to be back-propagated to lens design parameters [Na et al. 2024]. Similarly, *End-to-End Hybrid Refractive-Diffractive Lens Design with Differentiable Ray-Wave*

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*Model* combines ray tracing with wave propagation to jointly optimize refractive-diffractive optics and the reconstruction network [Yang et al. 2024]. These works suggest that optical simulation can be part of the design loop; in our case, future optimization could use the brightness, sharpness, separation from P1, and field-of-view coverage of P3/P4 as objectives for LED and camera placement.

In real captures, higher-order Purkinje images, especially P3 and P4, are relatively faint and sensitive to defocus and sensor noise. Related enhancement and adaptation methods suggest possible post-processing tools: unsupervised anti-noise computational imaging reconstructed  $64 \times 64$  images at 20 fps with only a 5% sampling ratio [Zhai et al. 2022], diffusion-based style transfer reduced synthetic-to-real FID to 54.56 compared with 65.31 and 73.26 for prior baselines [Chigot et al. 2025], and physics-informed diffusion reduced physical residuals by about two orders of magnitude in a Darcy-flow task and by 33%–35% in topology optimization [Bastek et al. 2025].

### 3 Background

#### 3.1 Optical basis of Purkinje images and VAC

In natural vision, vergence and accommodation are coupled: the eyes rotate toward a target while the crystalline lens changes optical power to focus it on the retina. In AR/VR displays, virtual depth cues may drive vergence to a distance different from the display focal plane, causing the vergence–accommodation conflict (VAC). As holographic and metasurface-waveguide AR displays move toward more realistic 3D imagery, measuring both vergence and accommodation becomes important for evaluating depth perception and visual comfort [Gopakumar et al. 2024].

Conventional eye trackers estimate gaze or vergence from pupil, iris, limbus, or corneal-glint features, which mainly describe eye rotation and provide limited access to accommodation [Nair et al. 2020]. Purkinje images offer a more direct optical cue: P1/P2 are mainly associated with the cornea, while P3/P4 originate from the crystalline lens surfaces and are related to accommodation [Lu et al. 2020][Ozhan et al. 2023]. Because P3/P4 are faint, easily defocused, and sensitive to LED–camera–eye geometry, we first study their formation in a controllable optical eye model before real IR capture.

#### 3.2 Simulation-to-capture pipeline for near-eye Purkinje imaging

A reliable Purkinje imaging pipeline requires the eye to be modeled as an optical system rather than an appearance object. Since P1–P4 are generated by reflection and refraction at corneal and lenticular surfaces, Arizona/ZEMAX-style schematic eye models are useful because they make accommodation, refractive interfaces, and ray paths physically controllable [Lu et al. 2020][Ozhan et al. 2023]. This also explains why eye-rendering methods often keep the eyeball or cornea as an explicit mesh: specular highlights, corneal refraction, and near-field glints are tied to surface geometry, while NeRF-like fields are more suitable for view synthesis, relighting, and periocular appearance modeling [???].

The same optical constraints determine capture quality. LED and camera placement must preserve a stable eye box while avoiding eyelid/eyelash occlusion, glint overlap, saturation, and defocus; otherwise weak P3/P4 reflections can merge with P1, fall outside

the field of view, or become unresolvable at the sensor resolution. Thus, the key background principle is that eye geometry, near-field illumination, camera focus/FOV, resolution, and reconstruction representation must be designed together before reliable Purkinje capture and later NeRF/3D Gaussian reconstruction can be expected [Kerbl et al. 2023]

### 4 Results and Discussions

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